

Adaptive Delayed-Acceptance MCMC with posterior guided active learning

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Abstract

Inverse uncertainty quantification tasks, such as Bayesian parameter estimation, are computationally demanding when the forward model is a physics-based numerical solver. In particular, for systems governed by partial differential equations, full-order discretizations, such as finite elements or finite volumes, make conventional Markov chain Monte Carlo sampling prohibitively expensive, due to the large number of forward evaluations required. Surrogate models can reduce this cost, but their effectiveness is often limited by the offline generation of high-fidelity training data. A natural remedy is to construct the surrogate online, concurrently with posterior sampling, so that training points concentrate in posterior-relevant regions. We propose an adaptive delayed-acceptance MCMC method in which a surrogate combining proper orthogonal decomposition and Gaussian process regression provides a cheap first-stage approximation of the likelihood. The surrogate is refined sequentially during sampling, and its predictive uncertainty is used to decide when a high-fidelity model evaluation is required. The delayed-acceptance construction preserves the high-fidelity posterior as the invariant target distribution, while reducing the number of expensive solver calls. Numerical experiments on two benchmark inverse problems, namely a mass-spring-damper system and an incompressible Navier–Stokes problem with variable geometry, show that the proposed strategy provides accurate posterior estimates with improved computational efficiency compared with standard single-stage MCMC and non-adaptive surrogate schemes.

Keywords: Bayesian inverse problems, Markov chain Monte Carlo, Gaussian process regression, Proper orthogonal decomposition, Surrogate modelling

1. Introduction

Parameter estimation is central to predictive modelling in engineering [1], where unknown physical or constitutive quantities must be inferred from indirect and often noisy observations. Reliable estimates are essential in applications such as fluid mechanics [2], climate modelling [3], structural engineering [4], and micro-scale devices [5], where parameter uncertainty can affect prediction quality, design decisions, and operational safety. Although high-fidelity numerical solvers have substantially improved the simulation of such systems, the underlying models are often governed by Partial Differential Equations (PDEs). As a result, each model evaluation may require the solution of a large-scale discretized problem. This cost becomes a major bottleneck in many-query settings, including uncertainty quantification and Bayesian inference, where thousands or millions of forward solves may be required [6, 7].

The Bayesian framework is particularly suited to this setting because it provides a systematic way to combine prior information, noisy observations, and complex forward models within a single probabilistic formulation [8, 9, 10]. Rather than returning only a point estimate, it characterizes the full posterior distribution of the unknown parameters, allowing one to quantify uncertainty, correlations, and credible regions. This is especially important for nonlinear or ill-posed inverse problems, where several parameter values may be consistent with the available data. Sampling-based methods, and Markov chain Monte Carlo (MCMC) in particular, are widely used in this context because they do not require explicit normalization of the posterior density and can be applied to a broad class of nonlinear and non-Gaussian models [11].

Their generality, however, comes at the cost of many likelihood evaluations. In PDE-constrained inverse problems, each such evaluation typically requires a high-fidelity forward solve, making conventional MCMC computationally demanding.

To address this cost, we develop an adaptive Surrogate Transition MCMC strategy based on a surrogate that combines Proper Orthogonal Decomposition (POD) and Gaussian Process (GP) regression. The surrogate is used as a cheap instrumental model for proposal screening, while high-fidelity corrections preserve the target posterior. The method uses an adaptive initial phase to refine the surrogate online, followed by a standard surrogate transition MCMC phase for posterior estimation.

Several alternative strategies have been developed to make Bayesian inference feasible for expensive forward models. Deterministic or approximate approaches, such as Laplace approximations [12, 13] and variational inference [14, 15], can reduce the cost by replacing posterior sampling with an optimization problem or a local approximation of the posterior. Sequential Monte Carlo [16, 17, 18] and ensemble methods [19, 20] provide alternative sampling-based strategies that can be effective in some settings. In this work, we focus on MCMC methods, which remain attractive because they can target the full posterior without imposing a parametric approximation, but whose cost is dominated by repeated likelihood evaluations.

Within MCMC, many acceleration strategies reduce the number or cost of high-fidelity forward solves by introducing a cheaper auxiliary model, either as a surrogate likelihood, a reduced-order approximation, or a hierarchy of models with different resolutions or fidelities. Surrogate transition method uses such an approximation to screen proposals before applying a high-fidelity correction, thereby preserving the target posterior while avoiding some expensive evaluations [21, 22]. Multilevel and multifidelity MCMC follow a related principle by exploiting hierarchies of models, meshes, or solvers, shifting part of the computational effort to cheaper approximations while retaining high-fidelity information to control bias or correct the sampling distribution [23]. The effectiveness of these methods depends strongly on the quality of the auxiliary model: if it is inaccurate in posterior-relevant regions, correction rates may be low, mixing may deteriorate, and the computational advantage can be limited.

A central question is therefore how to construct this auxiliary model. When a hierarchy of physical models or discretizations is available, coarse meshes or simplified solvers provide natural candidates. In many applications, however, such a hierarchy may be unavailable, difficult to calibrate, or insufficiently accurate in the posterior region. Surrogate and reduced-order models offer an alternative by approximating the parameter-to-observable map, or directly the likelihood, from a set of high-fidelity evaluations. Common choices include polynomial response surfaces, polynomial chaos expansions, projection-based reduced-order models such as reduced bases and POD, and Gaussian process emulators. Once trained, these models can be evaluated at much lower cost than the original solver.

This motivates an online strategy in which the surrogate is refined during sampling. Such adaptivity allows high-fidelity evaluations to be concentrated where the Markov chain actually explores, rather than over a fixed pre-selected design [24, 25]. At the same time, adaptive surrogate construction must be combined with the MCMC transition carefully: if the approximation changes inside a transition, the usual stationarity argument may fail [26, 27], and if the approximate likelihood is left uncorrected, the resulting chain may target a surrogate posterior rather than the desired high-fidelity posterior.

The method proposed in this work addresses these issues by combining three complementary ideas: reduced-order modelling, online surrogate learning, and surrogate transition MCMC. POD is used to compress the high-fidelity model outputs into a low-dimensional representation, so that the surrogate does not need to learn the full model response directly. GP regression is then used to learn the parameter dependence of the reduced coefficients and to provide a predictive uncertainty. This uncertainty enters the low-fidelity likelihood and is also used as an online learning indicator, triggering additional high-fidelity evaluations in regions where the surrogate is insufficiently resolved. Finally, the POD-GP approximation is embedded in a Surrogate Transition scheme: it provides a cheap first-stage likelihood, while the high-fidelity model is used in the correction stage to preserve the exact posterior target. After an adaptive tuning phase, the surrogate and sampling schedule are frozen, and posterior inference is performed using high-fidelity-corrected samples from the resulting production chain. A simplified figure representing the scheme is presented in Fig. 1.

The remainder of the paper is organized as follows. Section 2 presents a background part on the Bayesian

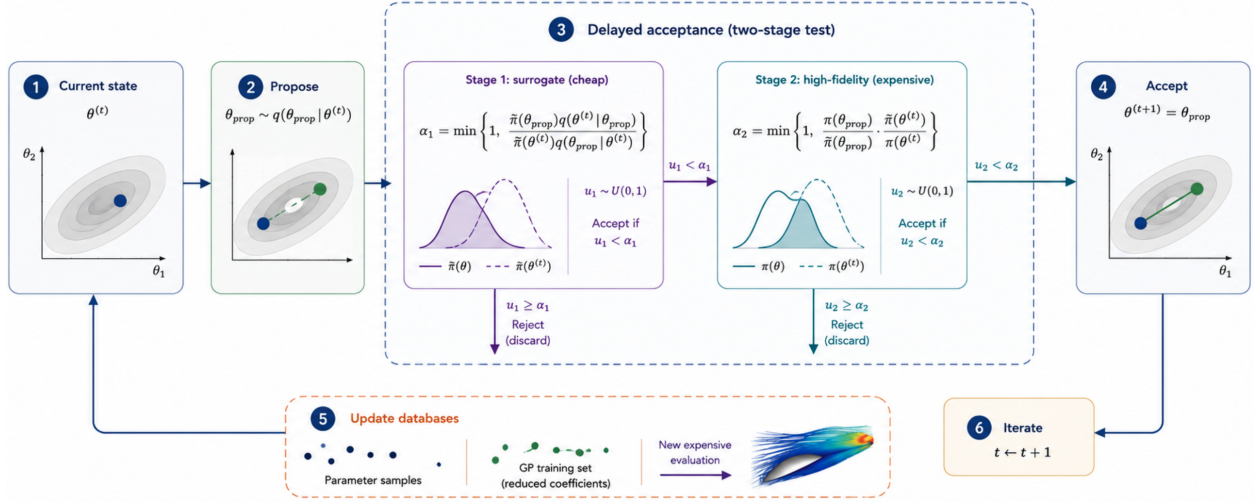


Figure 1: Placeholder: Simplified schematic of the proposed adaptive Surrogate Transition framework. The uncertainty-aware POD-GP surrogate is used to construct an approximate likelihood for inexpensive first-stage acceptance decisions. During the adaptive phase, posterior-guided acquisition identifies new high-fidelity evaluations to enrich the surrogate, while the Surrogate Transition subchain length is adjusted according to the estimated surrogate error. Once the surrogate satisfies the prescribed accuracy criterion, adaptation is terminated and the surrogate transition sampler proceeds with a fixed surrogate and transition kernel to generate posterior samples.

inverse problem and the delayed-acceptance construction. Section 3 presents the POD-GP surrogate and its introduction in the sampling scheme. Section 4 reports the numerical experiments on the benchmark problems. Finally, Section 5 summarizes the main findings and discusses possible extensions.

2. Background and Motivation

In this section, we briefly recall the Bayesian formulation of inverse problems and the use of MCMC methods for posterior inference. We then introduce surrogate-based delayed-acceptance strategies for reducing the cost of repeated evaluations of expensive forward models. This also highlights the main limitation of standard fixed-surrogate approaches, namely that their performance depends on the surrogate being accurate in the posterior-relevant region. These observations motivate the adaptive framework developed in the next section.

2.1. Bayesian inverse problem

Inverse problems are concerned with inferring unknown properties of a system from indirect measurements or observational data through a mathematical model. These unknowns are represented by a parameter vector $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^n$, while the available observations are denoted by $\mathbf{y}_{\text{obs}} \in \mathbb{R}^{d_y}$. We assume that parameters and observations are related through a high-fidelity forward map $\mathbf{f}_{\text{HF}} : \Theta \rightarrow \mathbb{R}^{d_y}$ and an additive noise term, so that

$$\mathbf{y}_{\text{obs}} = \mathbf{f}_{\text{HF}}(\boldsymbol{\theta}) + \boldsymbol{\varepsilon}. \quad (1)$$

In particular $\boldsymbol{\varepsilon}$ accounts for measurement noise and possible model discrepancy.

For a fixed $\boldsymbol{\theta}$, evaluating $\mathbf{f}_{\text{HF}}(\boldsymbol{\theta})$ defines the corresponding forward problem, which, in many applications, requires the (expensive) solution of a Partial Differential Equation (PDE). The resulting observables may represent sensor measurements or more general quantities of interest derived from the PDE solution, such as spatial averages, fluxes, or derivatives. In this setting, $\boldsymbol{\theta}$ typically collects model coefficients, boundary conditions, or initial conditions characterizing the underlying physical system.

In the Bayesian setting, the unknown parameters are treated as random variables. The goal is to characterize the posterior distribution $\pi(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}})$ of the parameters given the observations $\mathbf{y}_{\text{obs}}$. By Bayes' theorem,

$$\pi(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) = \frac{\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta})}{\pi(\mathbf{y}_{\text{obs}})}, \quad (2)$$

where $\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})$ denotes the likelihood of the data given the parameters, $\pi(\boldsymbol{\theta})$ is the prior distribution encoding the available knowledge on $\boldsymbol{\theta}$ before observing the data, and $\pi(\mathbf{y}_{\text{obs}})$ is the marginal distribution, or evidence, which acts as a normalizing constant and is given by

$$\pi(\mathbf{y}_{\text{obs}}) = \int_{\Theta} \pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) d\boldsymbol{\theta}. \quad (3)$$

The observation model induces a likelihood associated with the high-fidelity forward map. In general, if π_{ε} denotes the probability density of the additive error ε , then

$$\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = \pi_{\varepsilon}(\mathbf{y}_{\text{obs}} - \mathbf{f}_{\text{HF}}(\boldsymbol{\theta})). \quad (4)$$

For simplicity, in the following we assume additive Gaussian noise, $\varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{\Gamma}_{\text{obs}})$, although the proposed methodology is not restricted to this setting. Under this assumption, the high-fidelity likelihood takes the form

$$\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) \propto \exp\left(-\frac{1}{2} \|\mathbf{y}_{\text{obs}} - \mathbf{f}_{\text{HF}}(\boldsymbol{\theta})\|_{\mathbf{\Gamma}_{\text{obs}}^{-1}}^2\right). \quad (5)$$

The evidence integral in (3) cannot in general be evaluated analytically, and the posterior distribution is therefore not available in closed form. A standard strategy to explore such distributions is to resort to Markov chain Monte Carlo (MCMC) methods, which generate a Markov chain in the parameter space Θ having the posterior (2) as invariant distribution. Among the available choices, the Metropolis–Hastings (MH) algorithm is one of the most widely used. Starting from a current state $\boldsymbol{\theta}$, MH generates a candidate $\boldsymbol{\theta}'$ from a proposal distribution $q(\cdot \mid \boldsymbol{\theta})$ and accepts it with probability

$$\alpha(\boldsymbol{\theta}, \boldsymbol{\theta}') = \min\left\{1, \frac{\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}') \pi(\boldsymbol{\theta}') q(\boldsymbol{\theta} \mid \boldsymbol{\theta}')}{\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) q(\boldsymbol{\theta}' \mid \boldsymbol{\theta})}\right\}. \quad (6)$$

While this avoids the explicit computation of the evidence, each MH step still requires one evaluation of the high-fidelity likelihood, and hence one solve of the expensive forward model. The full procedure is summarized in Algorithm 1.

A comprehensive exploration of the posterior distribution typically requires a large number of MCMC iterations, as MH algorithm usually exhibits frequent rejections and strong correlation between successive samples, making the overall procedure computationally prohibitive. This motivates the introduction of surrogate-based strategies aimed at reducing the number of high-fidelity model evaluations while preserving the high-fidelity posterior as the target distribution.

2.2. Delayed-acceptance MCMC

In many applications, it is possible to replace the high fidelity model in (1) with a cheap approximation $\mathbf{f}_{\text{LF}} \approx \mathbf{f}_{\text{HF}}$. Replacing $\mathbf{f}_{\text{HF}}$ with $\mathbf{f}_{\text{LF}}$ in the observation model induces the low-fidelity likelihood

$$\pi_{\text{LF}}(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) = \pi_{\varepsilon}(\mathbf{y}_{\text{obs}} - \mathbf{f}_{\text{LF}}(\boldsymbol{\theta})), \quad (7)$$

which provides a computationally inexpensive approximation to the high-fidelity likelihood. When the surrogate model is sufficiently accurate, it may be used directly within an MH scheme to sample from an approximate posterior distribution. In general, however, constructing a sufficiently accurate surrogate over the entire parameter space may be computationally prohibitive, particularly for high-dimensional or strongly nonlinear forward models. Moreover, since the posterior often concentrates on a relatively small region of the parameter space, global surrogate accuracy is neither necessary nor necessarily the most efficient objective

Algorithm 1 Metropolis–Hastings (MH)

Input: Likelihood $\pi(\mathbf{y}_{\text{obs}} \mid \cdot)$, prior distribution $\pi(\cdot)$, proposal distribution $q(\cdot \mid \cdot)$, initial sample $\boldsymbol{\theta}^{(0)}$, number of samples N

Output: Chain of samples $\{\boldsymbol{\theta}^{(j)}\}_{j=1}^N$

- 1: **for** $j = 1, \dots, N$ **do**
- 2: Propose $\boldsymbol{\theta}' \sim q(\cdot \mid \boldsymbol{\theta}^{(j-1)})$
- 3: Compute the acceptance probability

$$\alpha(\boldsymbol{\theta}^{(j-1)}, \boldsymbol{\theta}') = \min \left\{ 1, \frac{\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}') \pi(\boldsymbol{\theta}') q(\boldsymbol{\theta}^{(j-1)} \mid \boldsymbol{\theta}')}{\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}^{(j-1)}) \pi(\boldsymbol{\theta}^{(j-1)}) q(\boldsymbol{\theta}' \mid \boldsymbol{\theta}^{(j-1)})} \right\}$$

- 4: Accept $\boldsymbol{\theta}'$ with probability α ; if accepted set $\boldsymbol{\theta}^{(j)} = \boldsymbol{\theta}'$, otherwise set $\boldsymbol{\theta}^{(j)} = \boldsymbol{\theta}^{(j-1)}$
 - 5: **end for**
-

Algorithm 2 Delayed-acceptance MCMC (DA)

Input: High-fidelity likelihood $\pi(\mathbf{y}_{\text{obs}} \mid \cdot)$, low-fidelity likelihood $\pi_{\text{LF}}(\mathbf{y}_{\text{obs}} \mid \cdot)$, prior distribution $\pi(\cdot)$, proposal distribution $q(\cdot \mid \cdot)$, initial sample $\boldsymbol{\theta}^{(0)}$, subchain length L , number of samples N

Output: Chain of samples $\{\boldsymbol{\theta}^{(j)}\}_{j=1}^N$

- 1: **for** $j = 1, \dots, N$ **do**
- 2: Starting from $\boldsymbol{\theta}^{(j-1)}$, generate an L -step low-fidelity MH subchain using Algorithm 1:

$$\{\boldsymbol{\vartheta}^{(\ell)}\}_{\ell=1}^L = \text{MH}\left(\pi_{\text{LF}}(\mathbf{y}_{\text{obs}} \mid \cdot), \pi(\cdot), q(\cdot \mid \cdot), \boldsymbol{\theta}^{(j-1)}, L\right)$$

- 3: Set the candidate $\boldsymbol{\theta}' = \boldsymbol{\vartheta}^{(L)}$
- 4: Compute the delayed-acceptance probability

$$\alpha_{\text{DA}}(\boldsymbol{\theta}^{(j-1)}, \boldsymbol{\theta}') = \min \left\{ 1, \frac{\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}') \pi_{\text{LF}}(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}^{(j-1)})}{\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}^{(j-1)}) \pi_{\text{LF}}(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}')} \right\}$$

- 5: Accept $\boldsymbol{\theta}'$ with probability α_{DA} ; if accepted set $\boldsymbol{\theta}^{(j)} = \boldsymbol{\theta}'$, otherwise set $\boldsymbol{\theta}^{(j)} = \boldsymbol{\theta}^{(j-1)}$
 - 6: **end for**
-

[24]. The errors in the surrogate forward model propagate through the likelihood and lead to discrepancies between the approximate and high-fidelity posterior distributions [28].

The idea underlying Delayed-Acceptance (DA) MCMC is to find a trade-off between computational efficiency of the approximation and accuracy of the high-fidelity target. In particular, the surrogate likelihood is used as an initial screening mechanism, and only proposals that pass this first stage are subsequently evaluated using the high-fidelity likelihood. In this way, expensive high-fidelity model evaluations are restricted to candidates that are deemed sufficiently promising by the surrogate.

More precisely, let $\pi_{\text{LF}}(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) \propto \pi_{\text{LF}}(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})\pi(\boldsymbol{\theta})$ denote the approximate posterior induced by the low-fidelity likelihood. Starting from the current state $\boldsymbol{\theta}$, a fixed-length MH subchain of L steps is first simulated targeting $\pi_{\text{LF}}(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}})$. The final state of this subchain, denoted by $\boldsymbol{\theta}'$, is then used as a candidate for the high-fidelity chain and is accepted with probability

$$\alpha_{\text{DA}}(\boldsymbol{\theta}, \boldsymbol{\theta}') = \min \left\{ 1, \frac{\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}') \pi_{\text{LF}}(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})}{\pi(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) \pi_{\text{LF}}(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}')} \right\}. \quad (8)$$

It is possible to show that, despite using the surrogate model to generate candidates, the resulting Markov chain preserves the high-fidelity posterior as its invariant distribution.

When the surrogate provides a sufficiently accurate representation of the high-fidelity target, DA can yield substantial computational gains by reducing the number of expensive model evaluations while improving mixing of the chain. In particular, the use of inexpensive low-fidelity subchains can increase the effective sample size obtained for a given computational budget. Conversely, when the surrogate poorly represents the high-fidelity posterior, candidates generated at the low-fidelity level may be frequently rejected by the high-fidelity correction, leading to low acceptance rates, increased self-correlation, and poor overall sampling efficiency. The performance of DA scheme therefore remains strongly dependent on the quality of the surrogate approximation.

This limitation has motivated adaptive approaches in which information acquired during sampling is used to improve the approximation in posterior-relevant regions or to correct discrepancies between low- and high-fidelity models [29]. The approach developed in this work follows this general motivation, but adopts a different strategy: the surrogate is refined online as sampling proceeds, while its predictive uncertainty is incorporated into the inference procedure to prevent unreliable surrogate predictions from influencing the exploration of the posterior.

3. Methodology

The proposed method embeds a probabilistic reduced-order surrogate within a delayed-acceptance MCMC framework. The low-fidelity model combines Proper Orthogonal Decomposition (POD) and Gaussian Process (GP) regression. POD provides a low-dimensional representation of the high-fidelity model output, whereas GP regression approximates the dependence of the reduced coefficients on the model parameters. The resulting POD-GP surrogate provides both a predictive mean and a predictive covariance in the observation space.

Compared with a standard delayed-acceptance scheme based on a deterministic surrogate, the probabilistic formulation offers two main advantages:

- **Uncertainty-aware likelihood:** the surrogate predictive covariance is incorporated into the low-fidelity likelihood by inflating the observation-noise covariance. The resulting likelihood therefore accounts not only for measurement uncertainty, but also for the uncertainty associated with the surrogate approximation. Ideally, predictions in poorly resolved regions are assigned a larger effective covariance, reducing the risk that surrogate errors drift the exploration of the parameter space.
- **Online surrogate enrichment:** the predictive uncertainty also provides a criterion for identifying parameter values at which the surrogate should be refined. Whenever the uncertainty at a sampled parameter exceeds a prescribed threshold, the high-fidelity model is evaluated and the resulting parameter-output pair is added to the training set. GP regression is particularly well suited to this sequential setting, since newly acquired observations can be used to update the approximation as sampling proceeds. In this way, high-fidelity evaluations generated during posterior exploration are reused to improve the surrogate in posterior-relevant regions, allowing the algorithm to rely progressively more on the low-fidelity model.

Surrogate adaptation is performed during an initial adaptive phase. In the same phase, the high-fidelity evaluations carried out at the correction stage of the delayed-acceptance algorithm are used to monitor the discrepancy between the low- and high-fidelity predictions. This discrepancy determines the length of the low-fidelity subchain: the subchain is shortened when the surrogate error is large and lengthened when the surrogate provides a sufficiently accurate approximation.

The adaptive phase is used exclusively to construct and calibrate the surrogate and to determine the correction schedule. In a second phase, both the surrogate and the subchain length are kept fixed. The resulting sampler is therefore a standard delayed-acceptance Markov chain with a fixed low-fidelity model and a high-fidelity correction step, and hence preserves the high-fidelity posterior as its invariant distribution.

3.1. Surrogate definition and approximate likelihood

The delayed-acceptance algorithm requires a cheap approximation of the high-fidelity parameter-to-observable map $\mathbf{f}_{\text{HF}} : \Theta \rightarrow \mathbb{R}^{d_y}$. To this end, we combine POD with GP regression. POD is first used to construct a low-dimensional representation of the model outputs, after which GP regression is employed to approximate the mapping between the parameter vector and the reduced coordinates. This substantially reduces the complexity of the regression problem while retaining the dominant features of the high-fidelity solutions.

3.1.1. Proper Orthogonal Decomposition

Let $\mathcal{D} = \left\{ \left(\boldsymbol{\theta}^{(i)}, \mathbf{f}_{\text{HF}}(\boldsymbol{\theta}^{(i)}) \right) \right\}_{i=1}^{N_s}$ denote an offline available dataset of input - (high fidelity) output pairs. Assuming the snapshots have been centered (i.e. removing the mean), the snapshot matrix is defined as

$$\mathbf{S} = \begin{bmatrix} \left| \mathbf{f}_{\text{HF}}(\boldsymbol{\theta}^{(1)}) \right. & \cdots & \left| \mathbf{f}_{\text{HF}}(\boldsymbol{\theta}^{(N_s)}) \right. \\ \hline \end{bmatrix} \in \mathbb{R}^{d_y \times N_s}. \quad (9)$$

We then perform the singular value decomposition of the snapshot matrix:

$$\mathbf{S} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top. \quad (10)$$

The resulting POD basis on the full grid is

$$\mathbf{\Phi} = [\mathbf{u}_1, \dots, \mathbf{u}_r], \quad (11)$$

which is formed by retaining the first r left singular vectors, where r is chosen such that

$$\frac{\sum_{i=1}^r \sigma_i^2}{\sum_i \sigma_i^2} \geq 0.95. \quad (12)$$

For a new high fidelity solution we can then obtain its reduced dimension representation through the projection operator $\mathbf{\Phi}$:

$$\mathbf{c}(\boldsymbol{\theta}) = \mathbf{\Phi}^\top \mathbf{f}_{\text{HF}}(\boldsymbol{\theta}). \quad (13)$$

At the same time, given the reduced representation, the high fidelity solution is approximated by re-projecting onto the full space: The high-fidelity solution is approximated by

$$\mathbf{f}_{\text{HF}}(\boldsymbol{\theta}) \approx \mathbf{\Phi} \mathbf{c}(\boldsymbol{\theta}). \quad (14)$$

The remaining task is therefore to approximate the nonlinear mapping $\boldsymbol{\theta} \mapsto \mathbf{c}(\boldsymbol{\theta})$, which is achieved using Gaussian Process regression.

3.1.2. Gaussian Process Regression

Let $\mathcal{D}_{\text{GP}} = \left\{ \left(\boldsymbol{\theta}^{(i)}, \mathbf{c}^{(i)} \right) \right\}_{i=1}^{N_{\text{GP}}}$, denote the available training set, where the reduced coefficients $\mathbf{c}^{(i)}$ are obtained by projecting the corresponding high-fidelity solutions onto the POD basis. Note that $N_{\text{GP}} \neq N_s$ in general and we will describe in the overall algorithm how these data are obtained. Each POD coefficient is approximated independently using GP regression. For the j -th coefficient, we assume the prior

$$c_j(\boldsymbol{\theta}) \sim \mathcal{GP}(0, k_{\mathbf{w}}(\boldsymbol{\theta}, \boldsymbol{\theta}')), \quad (15)$$

where $k_{\mathbf{w}}$ is a parametric covariance kernel with hyperparameters $\mathbf{w}$. In this work, standard kernels such as the squared exponential or Matérn covariance functions may be employed. Conditioning on $\mathcal{D}_{\text{GP}}$, the predictive distribution at $\boldsymbol{\theta}$ is

$$c_j(\boldsymbol{\theta}) \mid \mathcal{D}_{\text{GP}} \sim \mathcal{N}(\hat{c}_j(\boldsymbol{\theta}), s_j^2(\boldsymbol{\theta})), \quad (16)$$

with predictive mean and variance

$$\hat{c}_j(\boldsymbol{\theta}) = \mathbf{k}_{\mathbf{w}}(\boldsymbol{\theta})^\top \mathbf{K}_{\mathbf{w}}^{-1} \mathbf{c}_j, \quad (17)$$

$$s_j^2(\boldsymbol{\theta}) = k_{\mathbf{w}}(\boldsymbol{\theta}, \boldsymbol{\theta}) - \mathbf{k}_{\mathbf{w}}(\boldsymbol{\theta})^\top \mathbf{K}_{\mathbf{w}}^{-1} \mathbf{k}_{\mathbf{w}}(\boldsymbol{\theta}). \quad (18)$$

Here, $\mathbf{K}_{\mathbf{w}} \in \mathbb{R}^{N_{\text{GP}} \times N_{\text{GP}}}$ is the covariance matrix with entries

$$[\mathbf{K}_{\mathbf{w}}]_{mn} = k_{\mathbf{w}}(\boldsymbol{\theta}^{(m)}, \boldsymbol{\theta}^{(n)}), \quad (19)$$

and

$$\mathbf{k}_{\mathbf{w}}(\boldsymbol{\theta}) = \left[k_{\mathbf{w}}(\boldsymbol{\theta}, \boldsymbol{\theta}^{(1)}), \dots, k_{\mathbf{w}}(\boldsymbol{\theta}, \boldsymbol{\theta}^{(N_{\text{GP}})}) \right]^\top. \quad (20)$$

Moreover,

$$\mathbf{c}_j = \left[c_j^{(1)}, \dots, c_j^{(N_{\text{GP}})} \right]^\top \quad (21)$$

collects the training values of the j -th POD coefficient.

Since the POD basis decorrelates the reduced coefficients, each coefficient is modeled independently by a scalar Gaussian process. This avoids the need for a multi-output Gaussian process while retaining the dominant variability of the solution. We thus have:

$$\hat{\mathbf{c}}(\boldsymbol{\theta}) = [\hat{c}_1(\boldsymbol{\theta}), \dots, \hat{c}_r(\boldsymbol{\theta})]^\top, \quad (22)$$

while the corresponding covariance matrix is

$$\boldsymbol{\Sigma}_c(\boldsymbol{\theta}) = \text{diag}(s_1^2(\boldsymbol{\theta}), \dots, s_r^2(\boldsymbol{\theta})). \quad (23)$$

The surrogate prediction of the high-fidelity model, comprising both the mean and the predictive variance, is therefore obtained by projecting the two reduced quantities on the full space:

$$\mathbf{f}_{\text{LF}}(\boldsymbol{\theta}) = \boldsymbol{\Phi} \hat{\mathbf{c}}(\boldsymbol{\theta}) \in \mathbb{R}^{d_y}, \quad \boldsymbol{\Sigma}_{\text{LF}}(\boldsymbol{\theta}) = \boldsymbol{\Phi} \boldsymbol{\Sigma}_c(\boldsymbol{\theta}) \boldsymbol{\Phi}^\top \in \mathbb{R}^{d_y \times d_y} \quad (24)$$

Assuming additive Gaussian observation noise and independent Gaussian surrogate errors, the predictive distribution of the observations remains Gaussian, with covariance given by the sum of the observation covariance $\boldsymbol{\Gamma}_{\text{obs}}$ and the surrogate predictive covariance $\boldsymbol{\Sigma}_{\text{LF}}(\boldsymbol{\theta})$. The resulting approximate likelihood is

$$\pi_{\text{LF}}(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}) \propto |\boldsymbol{\Gamma}_{\text{obs}} + \boldsymbol{\Sigma}_{\text{LF}}(\boldsymbol{\theta})|^{-1/2} \exp\left(-\frac{1}{2} \|\mathbf{y}_{\text{obs}} - \mathbf{f}_{\text{LF}}(\boldsymbol{\theta})\|_{(\boldsymbol{\Gamma}_{\text{obs}} + \boldsymbol{\Sigma}_{\text{LF}}(\boldsymbol{\theta}))^{-1}}^2\right). \quad (25)$$

More generally, for non-Gaussian likelihoods, the surrogate uncertainty can be incorporated by marginalizing the likelihood with respect to the surrogate predictive distribution. An simplified scheme of the surrogate construction and usage is presented in Fig. 2.

3.2. Adaptive sampling scheme

The surrogate model introduced above is embedded within an adaptive delayed-acceptance framework consisting of three stages:

- **Offline stage:** an initial training set is constructed by drawing N_s parameter samples from the prior distribution using a space-filling design, such as Latin Hypercube Sampling. The corresponding high-fidelity evaluations are used to construct the POD basis $\boldsymbol{\Phi}$ and to initialize the GP training set $\mathcal{D}_{\text{GP}}^{(0)}$, yielding the initial surrogate $\mathbf{f}_{\text{LF}}^{(0)}$ and the corresponding approximate likelihood $\pi_{\text{LF}}^{(0)}$.

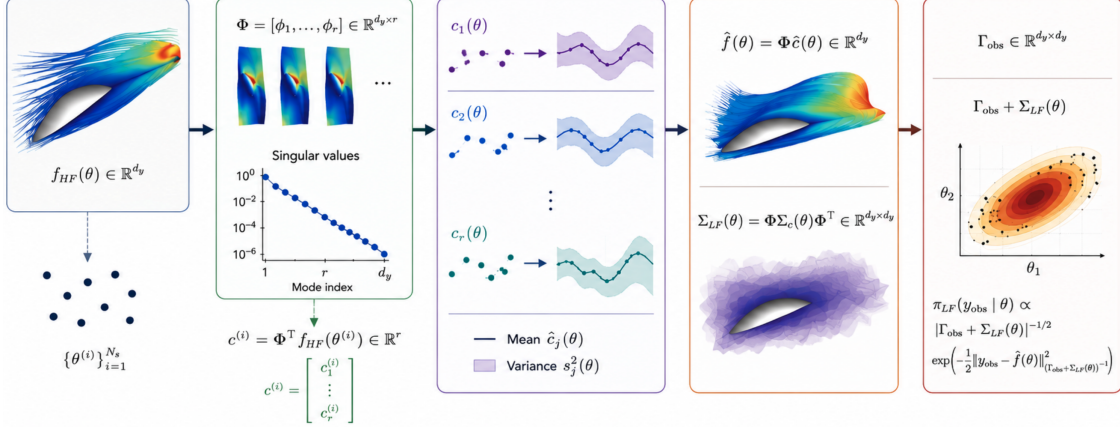


Figure 2: GPT-Place holder. Simplified schematic of the LF model used to construct the approximate likelihood. The surrogate combines POD for dimensionality reduction with GP regression to predict the reduced coefficients. The GP predictive uncertainty is first estimated in the reduced space and then propagated to the full observation space, and is then incorporated into the approximate likelihood.

- **Adaptive online stage:** posterior sampling is performed using delayed-acceptance MCMC initialized with the surrogate likelihood $\pi_{LF}^{(0)}$ and an initial subchain length $L^{(0)}$. During this stage, the surrogate is iteratively enriched by incorporating new high-fidelity evaluations selected according to the adaptive strategy described below. The subchain length is updated simultaneously to account for the evolving surrogate accuracy. At adaptive iteration j , the algorithm is characterized by the GP training set $\mathcal{D}_{GP}^{(j)}$, the surrogate $\mathbf{f}_{LF}^{(j)}$, the approximate likelihood $\pi_{LF}^{(j)}$, and the subchain length $L^{(j)}$. The adaptation is performed for at most N_{adapt} iterations or until the prescribed stopping criterion is satisfied.
- **Fixed-kernel online stage:** once adaptation is terminated, both the surrogate and the subchain length are frozen. The delayed-acceptance sampler is then continued using the fixed approximate likelihood $\pi_{LF}^{(N_{adapt})}$ and subchain length $L^{(N_{adapt})}$, thereby defining a fixed transition kernel for posterior sampling and guaranteeing sampling from the true posterior.

For what concerns the surrogate likelihood, at a candidate parameter value θ , the surrogate provides the predictive mean $\mathbf{f}_{LF}^{(j)}(\theta)$ and predictive variance $\sigma_{LF}^{2,(j)}(\theta)$. A high-fidelity evaluation is triggered whenever the average predictive variance satisfies

$$\frac{1}{d_y} \sum_{i=1}^{d_y} \sigma_{LF,i}^{2,(j)}(\theta) > \gamma^2, \quad (26)$$

where γ is a prescribed uncertainty threshold. The resulting pair $(\theta, \mathbf{f}_{HF}(\theta))$ is added to $\mathcal{D}_{GP}^{(j)}$, and the GP surrogate is updated. High-fidelity evaluations required by the second delayed-acceptance stage are also incorporated into the training set in the same manner.

The subchain length is adapted using the discrepancy between the surrogate prediction, evaluated before enrichment, and the corresponding high-fidelity output. Specifically, at the k -th high-fidelity evaluation, the error is measured as

$$e_k = \left[\frac{1}{d_y} \left\| \mathbf{f}_{HF}(\theta_k) - \mathbf{f}_{LF}^{(j)}(\theta_k) \right\|_2^2 \right]^{1/2}. \quad (27)$$

After every N_L high-fidelity evaluations, the subchain length is updated according to

$$L^{(j+1)} = \begin{cases} \max\{L_{\min}, \lfloor \rho_- L^{(j)} \rfloor\}, & e_k > \varepsilon_L, \\ \min\{L_{\max}, \lceil \rho_+ L^{(j)} \rceil\}, & e_k \leq \varepsilon_L, \end{cases} \quad (28)$$

Algorithm 3 Adaptive delayed-acceptance sampling scheme

Input: Initial design size N_s , initial subchain length $L^{(0)}$, maximum number of adaptive blocks N_{adapt} , adaptive block size N_b , and final sampling length N_{final}

Output: Fixed-kernel posterior chain $\{\boldsymbol{\theta}^{(n)}\}_{n=1}^{N_{\text{final}}}$

Offline Stage:

- 1: Draw $\{\boldsymbol{\theta}^{(i)}\}_{i=1}^{N_s}$ from the prior using a space-filling design and evaluate $\mathbf{f}_{\text{HF}}$
- 2: Construct the POD basis Φ and initialize $\mathcal{D}_{\text{GP}}^{(0)}$
- 3: Train the initial surrogate and define $\pi_{\text{LF}}^{(0)}$

Adaptive Online Stage:

- 4: Initialize the chain state $\boldsymbol{\theta}^{(0)}$ and set $j = 0$
- 5: **while** $j < N_{\text{adapt}}$ and the adaptive stopping criterion is not satisfied **do**
- 6: Run an N_b -step delayed-acceptance block using Algorithm 2, with fixed $\pi_{\text{LF}}^{(j)}$ and $L^{(j)}$
- 7: Collect the high-fidelity evaluations generated by the delayed-acceptance corrections
- 8: Select additional enrichment points for which

$$\frac{1}{d_y} \text{tr}(\Sigma_{\text{LF}}^{(j)}(\boldsymbol{\theta})) > \gamma^2$$

and evaluate $\mathbf{f}_{\text{HF}}$ at these points

- 9: Form $\mathcal{D}_{\text{GP}}^{(j+1)}$ by augmenting $\mathcal{D}_{\text{GP}}^{(j)}$ with the newly acquired data
- 10: Compute the surrogate discrepancies at the newly evaluated high-fidelity points
- 11: Retrain the GP surrogate and define $\pi_{\text{LF}}^{(j+1)}$
- 12: Update $L^{(j+1)}$ according to the error-based adaptation rule
- 13: Set $j \leftarrow j + 1$
- 14: **end while**

Fixed Kernel Online Stage:

- 15: Freeze the surrogate likelihood $\pi_{\text{LF}}^* = \pi_{\text{LF}}^{(j)}$ and the subchain length $L^* = L^{(j)}$
- 16: Generate the final chain by applying Algorithm 2 with π_{LF}^* , L^* , and $N = N_{\text{final}}$:

$$\left\{ \boldsymbol{\theta}^{(n)} \right\}_{n=1}^{N_{\text{final}}} = \text{DA} \left(\pi(\mathbf{y}_{\text{obs}} \mid \cdot), \pi_{\text{LF}}^*(\mathbf{y}_{\text{obs}} \mid \cdot), \pi(\cdot), q(\cdot \mid \cdot), \boldsymbol{\theta}^{(j-1)}, L^*, N_{\text{final}} \right)$$

where ε_L is the target surrogate error, $0 < \rho_- < 1$ and $\rho_+ > 1$ are contraction and expansion factors, and $L_{\min}$ and $L_{\max}$ prescribe admissible bounds. Hence, inaccurate surrogate predictions lead to more frequent high-fidelity corrections, whereas sufficiently accurate predictions allow longer surrogate subchains.

The adaptive stage is terminated when the surrogate error remains below the prescribed tolerance for a specified number of consecutive updates, or when the maximum number of adaptive iterations, N_{adapt} , is reached. The surrogate model and the subchain length obtained at termination are then fixed for the final delayed-acceptance sampling stage.

4. Results

5. Conclusion

References

- [1] S. L. Brunton, J. N. Kutz, Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control, Cambridge University Press, 2019.

- [2] S. L. Cotter, M. Dashti, J. C. Robinson, A. M. Stuart, Bayesian inverse problems for functions and applications to fluid mechanics, *Inverse Problems* 25 (11) (2009) 115008. doi:10.1088/0266-5611/25/11/115008.
URL <https://doi.org/10.1088/0266-5611/25/11/115008>
- [3] T. Schneider, S. Lan, A. Stuart, J. Teixeira, Earth system modeling 2.0: A blueprint for models that learn from observations and targeted high-resolution simulations, *Geophysical Research Letters* 44 (24) (2017) 12,396–12,417. arXiv:<https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1002/2017GL076101>, doi:<https://doi.org/10.1002/2017GL076101>.
URL <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2017GL076101>
- [4] A. Gallet, S. Rigby, T. N. Tallman, X. Kong, I. Hajirasouliha, A. Liew, D. Liu, L. Chen, A. Hauptmann, D. Smyl, Structural engineering from an inverse problems perspective, *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 478 (2257) (2022) 20210526. arXiv:<https://royalsocietypublishing.org/rspa/article-pdf/doi/10.1098/rspa.2021.0526/355952/rspa.2021.0526.pdf>, doi:10.1098/rspa.2021.0526.
URL <https://doi.org/10.1098/rspa.2021.0526>
- [5] F. Zacchei, F. Rizzini, G. Gattere, A. Frangi, A. Manzoni, Neural networks based surrogate modeling for efficient uncertainty quantification and calibration of mems accelerometers, *International Journal of Non-Linear Mechanics* 167 (2024) 104902. doi:<https://doi.org/10.1016/j.ijnonlinmec.2024.104902>.
URL <https://www.sciencedirect.com/science/article/pii/S0020746224002671>
- [6] T. J. Sullivan, *Introduction to uncertainty quantification*, Vol. 63, Springer, 2015.
- [7] A. M. Stuart, Inverse problems: A bayesian perspective, *Acta Numerica* 19 (2010) 451–559. doi:10.1017/S0962492910000061.
- [8] A. Tarantola, *Inverse problem theory and methods for model parameter estimation*, SIAM, 2005.
- [9] J. P. Kaipio, E. Somersalo, *Statistical and computational inverse problems*, Springer, 2005.
- [10] R. C. Smith, *Uncertainty quantification: theory, implementation, and applications*, SIAM, 2024.
- [11] C. P. Robert, G. Casella, *Monte Carlo statistical methods*, Vol. 2, Springer, 2004.
- [12] C. Schillings, B. Sprungk, P. Wacker, On the convergence of the laplace approximation and noise-level-robustness of laplace-based monte carlo methods for bayesian inverse problems, *Numerische Mathematik* 145 (4) (2020) 915–971. doi:10.1007/s00211-020-01131-1.
URL <https://doi.org/10.1007/s00211-020-01131-1>
- [13] T. Helin, R. Kretschmann, Non-asymptotic error estimates for the laplace approximation in bayesian inverse problems, *Numerische Mathematik* 150 (2) (2022) 521–549. doi:10.1007/s00211-021-01266-9.
URL <https://doi.org/10.1007/s00211-021-01266-9>
- [14] J. Povala, I. Kazlauskaitė, E. Febrianto, F. Cirak, M. Girolami, Variational bayesian approximation of inverse problems using sparse precision matrices, *Computer Methods in Applied Mechanics and Engineering* 393 (2022) 114712. doi:<https://doi.org/10.1016/j.cma.2022.114712>.
URL <https://www.sciencedirect.com/science/article/pii/S0045782522000822>
- [15] J. Jia, Q. Zhao, Z. Xu, D. Meng, Y. Leung, Variational bayes’ method for functions with applications to some inverse problems, *SIAM Journal on Scientific Computing* 43 (1) (2021) A355–A383. arXiv:<https://doi.org/10.1137/19M130409X>, doi:10.1137/19M130409X.
URL <https://doi.org/10.1137/19M130409X>

- [16] A. Beskos, A. Jasra, E. A. Muzaffer, A. M. Stuart, Sequential monte carlo methods for bayesian elliptic inverse problems, *Statistics and Computing* 25 (4) (2015) 727–737. doi:10.1007/s11222-015-9556-7. URL <https://doi.org/10.1007/s11222-015-9556-7>
- [17] N. Kantas, A. Beskos, A. Jasra, Sequential monte carlo methods for high-dimensional inverse problems: A case study for the navier–stokes equations, *SIAM/ASA Journal on Uncertainty Quantification* 2 (1) (2014) 464–489. doi:10.1137/130930364.
- [18] J. Wan, N. Zabaras, A bayesian approach to multiscale inverse problems using the sequential monte carlo method, *Inverse Problems* 27 (10) (2011) 105004. doi:10.1088/0266-5611/27/10/105004. URL <https://doi.org/10.1088/0266-5611/27/10/105004>
- [19] M. A. Iglesias, K. J. H. Law, A. M. Stuart, Ensemble kalman methods for inverse problems, *Inverse Problems* 29 (4) (2013) 045001. doi:10.1088/0266-5611/29/4/045001.
- [20] M. A. Iglesias, A regularizing iterative ensemble kalman method for pde-constrained inverse problems, *Inverse Problems* 32 (2) (2016) 025002. doi:10.1088/0266-5611/32/2/025002.
- [21] J. A. Christen, C. Fox, Markov chain monte carlo using an approximation, *Journal of Computational and Graphical Statistics* 14 (4) (2005) 795–810. doi:10.1198/106186005X76983.
- [22] M. Banterle, C. Grazian, A. Lee, C. P. Robert, Accelerating metropolis–hastings algorithms by delayed acceptance, *Foundations of Data Science* 1 (2) (2019) 103–128. doi:10.3934/fods.2019005.
- [23] B. Peherstorfer, K. Willcox, M. Gunzburger, Survey of multifidelity methods in uncertainty propagation, inference, and optimization, *SIAM Review* 60 (3) (2018) 550–591. doi:10.1137/16M1082469.
- [24] J. Li, Y. M. Marzouk, Adaptive construction of surrogates for the bayesian solution of inverse problems, *SIAM Journal on Scientific Computing* 36 (3) (2014) A1163–A1186. doi:10.1137/130938189.
- [25] P. R. Conrad, Y. M. Marzouk, N. S. Pillai, A. Smith, Accelerating asymptotically exact MCMC for computationally intensive models via local approximations, *Journal of the American Statistical Association* 111 (516) (2016) 1591–1607. doi:10.1080/01621459.2015.1096787.
- [26] G. O. Roberts, J. S. Rosenthal, Coupling and ergodicity of adaptive markov chain monte carlo algorithms, *Journal of Applied Probability* 44 (2) (2007) 458–475. doi:10.1239/jap/1183667414.
- [27] C. Andrieu, J. Thoms, A tutorial on adaptive MCMC, *Statistics and Computing* 18 (4) (2008) 343–373. doi:10.1007/s11222-008-9110-y.
- [28] S. L. Cotter, M. Dashti, A. M. Stuart, Approximation of bayesian inverse problems for pdes, *SIAM Journal on Numerical Analysis* 48 (1) (2010) 322–345. doi:10.1137/090770734.
- [29] M. B. Lykkegaard, T. J. Dodwell, C. Fox, G. Mingas, R. Scheichl, Multilevel delayed acceptance mcmc, *SIAM/ASA Journal on Uncertainty Quantification* 11 (1) (2023) 1–30. doi:10.1137/22M1476770.